

Business Understanding

The Quest
The Data



<https://github.com/FIIT-IAU>

Data Understanding

Data Visualization
Histogram
Distribution plot
Q-Q plot
Boxplot
Regplot
Scatterplot
Pairplot
Pearson correlation plot
Violin plot
Heatmap
Probability Density Functions
Data Distribution
Gaussian and other distributions
Measure of Central Tendency
Mean
Median
Mode
Variance
Standard deviation
Skewness
Kurtosis
Statistical Testing
Shapiro-Wilk test
Kolmogorov-Smirnov test
Anderson-Darling test
Levene test
Student's T-test
Analysis of Variance
Mann-Whitney U test
Kruskal-Wallis H test
Degrees of freedom
Confidence level
Statistical power
Effect size
Correlation does not imply causation
Communication and Soft skills

Data Preparation

Data Cleaning
Outlier detection
DBScan Clustering
Isolation Forest
Imputation of missing values
kNN
Linear column removing
Deduplication
Data Integration
Data Transformations
Data normalization
Data standardization
Data scaling
Transformers
Power Transformer
Quantile Transformer
Encoding
Ordinal Encoding
One Hot Encoding
Binary Encoding
BaseN Encoding
Hashing (MD5)
Target Encoding
Leave One Out
Data discretization
Invert transforms
Data Preprocessing
Feature Extraction
Bag-of-Words
IF-IDF
Word Embedding
Distributional Semantics
Contextual Representation
Feature Encoding
Feature Selection
Filter
Variance Threshold
Mutual Information
Chi-Squared (χ^2)
F-value
Wrapper
Recursive Feature Elimination
Selection from model
Embedding methods
LASSO
Ridge
Elastic Net
Dimensionality Reduction
Matrix Factorization
Singular Value Decomposition
Principal Components Analysis
Model-based projection
Linear Discriminant Analysis
Manifold learning
Multi-Dimensional Scaling
Isomap
t-SNE

Modeling

Machine Learning
Linear Regression
Logistic Regression
Multinomial Naïve Bayes
Decision Trees
Random Forest
Ensemble Learning
Stacking/Voting
Bagging
Boosting
Deep Learning
Convolutional Neural Networks
Recurrent Neural Networks

Evaluation

Learning System Design
Quantify Model Performance
Classification Metrics
Regression Metrics
Generalization in Machine Learning
Overfitting
Underfitting
Bias, Variance and their trade-off
Regularizations
Evaluation Techniques
Cross-validation
Hold-out
k-fold
Time series datasets
Imbalanced datasets
Undersampling
Oversampling
Model Selection
Hyperparameter Tuning
Grid Search
Random Search
Bayesian Optimization
Optimization
Nature-Inspired methods for feature selection
Gradient Descent for Neural Networks
Other Optimizations
Communication and Soft skills

Deployment
Production

Data Curation and Model Curation
Responsible AI